

人工智能专业 本科

人工智能

Artificial Intelligence (AI)

2023-2024

know yourself



About Course

❖ Course website

[*http://www.knowyourself.xyz/AI/*](http://www.knowyourself.xyz/AI/)

- Lecture slides

- Project solutions

- Questions

- Other resources

人工智能 Artificial Intelligence 登录 | 注册 设为首页 帮助 BBS讨论区

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[问题解答](#) [意见建议](#)

首页

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新闻公告

课程学习

练习题

资源共享

问题与建议

实验

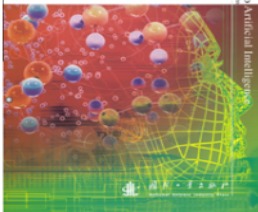
成果展示

友情链接

北京理工大学

人工智能网站

任课教师



教材介绍

刘峡壁, 人工智能导论-方法与系统, 国防工业出版社, 2008年08月.

本书以人工智能的哲学基础和工程实践为主线, 系统全面地介绍人工智能的核心知识和最新进展。目的在于帮助读者建立起对于人工智能的总体认识, 为以后进入人工智能各分支的研究和应用奠定基础。

哲学基础涵盖目前实现人工智能的六大类思想和方法, 分别是 (1) 符号主义, 分为基于搜索的问题求解和基于知识的推理两个主题; (2) 连接主义, 即人工神经网络; (3) 学习主义, 指机器学习方法; (4) 行为主义, 含智能体和强化学习; (5) 进化主义, 指遗传算法、进化规划、进化策略这三种进化计算方法; (6) 群体主义, 分为多智能体系统和群智能优化方法两个部分。群智能优化方法包括蚁群优化算法和粒子群优化算法。工程实践涉及具有代表性和广泛应用性的人工智能系统—专家系统, 以及实现人工智能的软硬件条件, 包括两种智能程序设计语言 (LISP、PROLOG) 和三种下一代智能计算机 (光计算机、量子计算机、生物计算机)。

除了这些主要内容以外, 本书还有两项特色性的辅助内容: (1) 在每一章最后推荐了深入学习的资源, 为希望更加透彻地理解和思考有关问题的读者提供获取相应信息的地图。 (2) 在附录部分提供了汉英-英汉术语对照与索引, 有助于读者形成人工智能的英文概念体系, 获得查询和阅读有关英文文献的初步能力。

本书可作为相关专业本科生和研究生的人工智能教材, 也可供从事人工智能研究和应用的专业技术人员参考。

Grading

Final exam: 50%

Team projects: 40%

Participation in the class and Exercises: 10%

Bonus: at most 12 scores

Course Overview

PartI Machine Learning (new book: chapters 2-5)

Concepts, Methods, Supervised and Unsupervised Learning

Part II Connectionism (new book: chapters 9-11)

Concepts, Problems, Models

PartII Symbolic AI (old book: chapters 2-3)

Problem representation, Graph Search, Adversarial Search, Reasoning

PartV Evolutionary Computation (old book: chapters 7)

Genetic Algorithms, Evolutionary Programming, Evolutionary Strategies

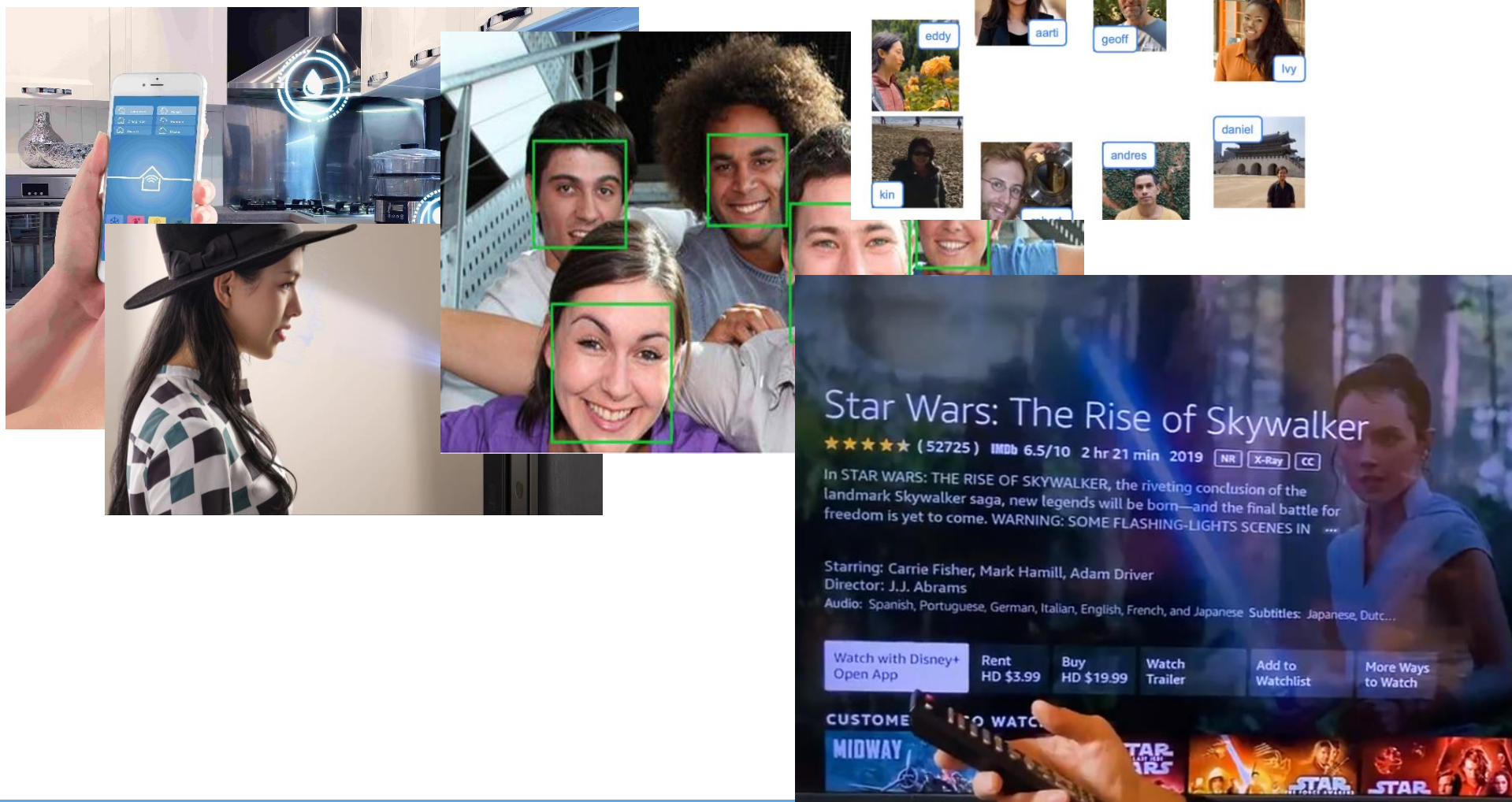
PartVI Computational Swarm Intelligence (old book: chapters 8)

Ant Colony Optimization, Particle Swarm Optimization

PartIV Nouvelle AI (old book: chapters 6, new book: chapter 8)

Agent, Reinforcement Learning

❖ AI in your life



Intro



[Segment Anything | Meta AI \(segment-anything.com\)](https://segment-anything.com/)

Intro

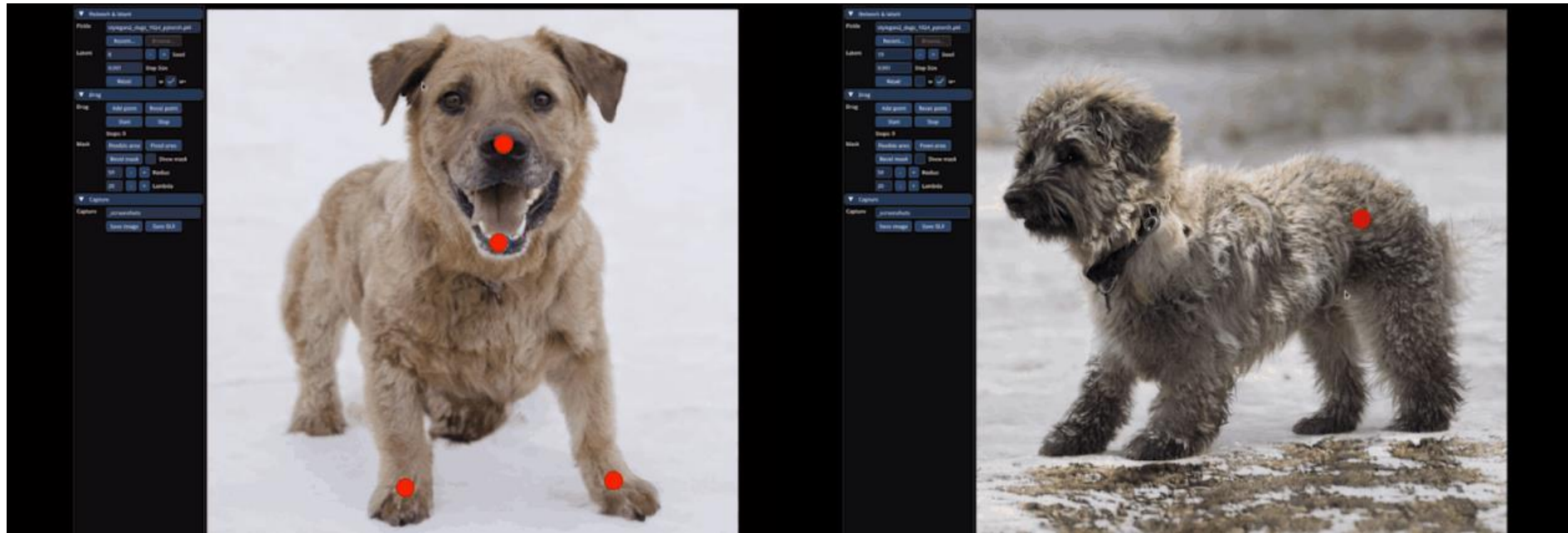


Massachusetts
Institute of
Technology

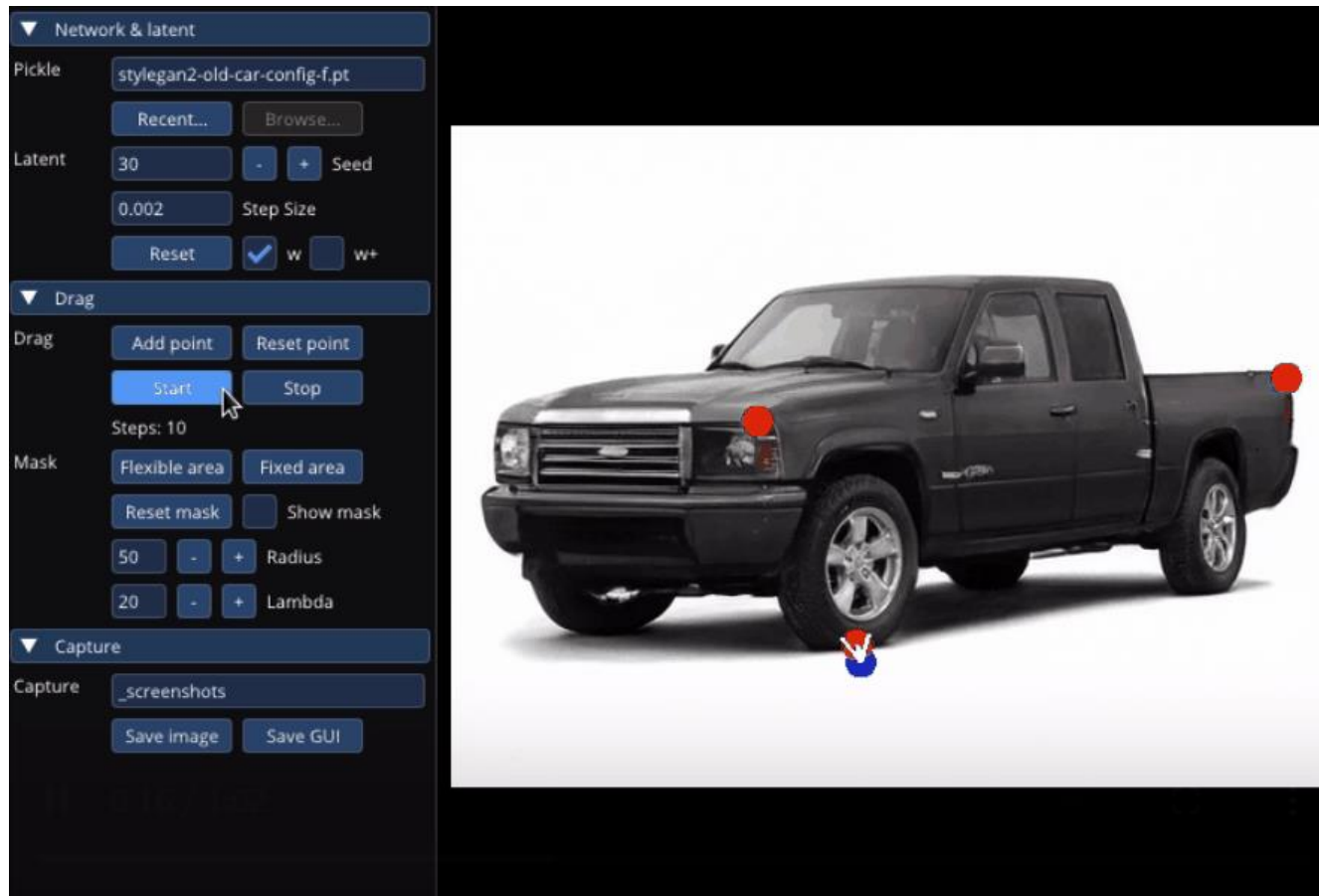


Drag Your GAN: Interactive Point-based Manipulation on the Generative Image Manifold

Xingang Pan^{1,2} Ayush Tewari³ Thomas Leimkühler¹ Lingjie Liu^{1,4} Abhimitra Meka⁵ Christian Theobalt^{1,2}



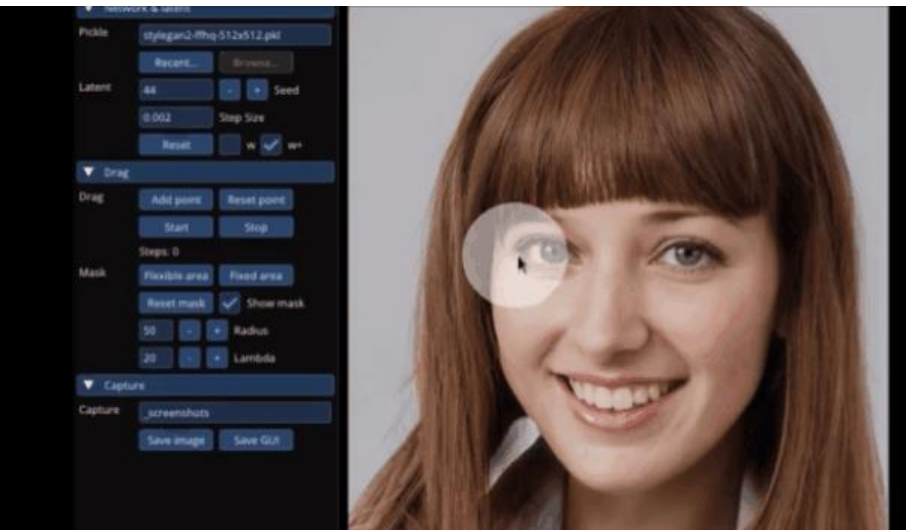
Intro



Intro



Intro



AK  @_akhaliq · May 19

Drag Your GAN: Interactive Point-based Manipulation on the Generative Image Manifold

paper page: huggingface.co/papers/2305.10...



337

7,060

21.7K

2.9M





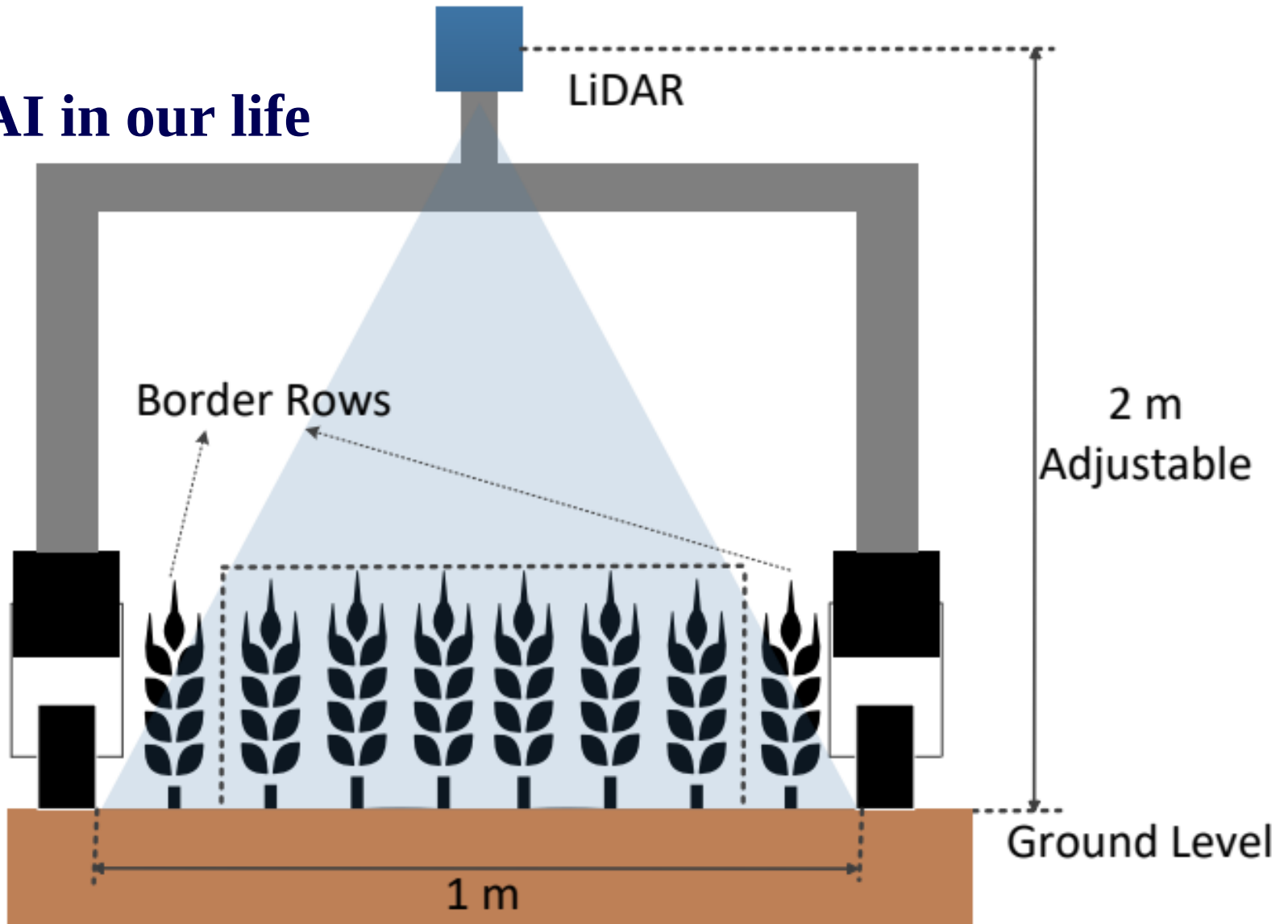
Biomass Prediction with 3D Point Clouds from LiDAR

Liyuan Pan^{1,2,*}, Liu Liu², Anthony G. Condon¹, Gonzalo M. Estavillo¹, Robert A. Coe^{1,3}, Geoff Bull^{1,3}
Eric A. Stone^{1,2}, Lars Petersson⁴, Vivien Rolland^{1,*}

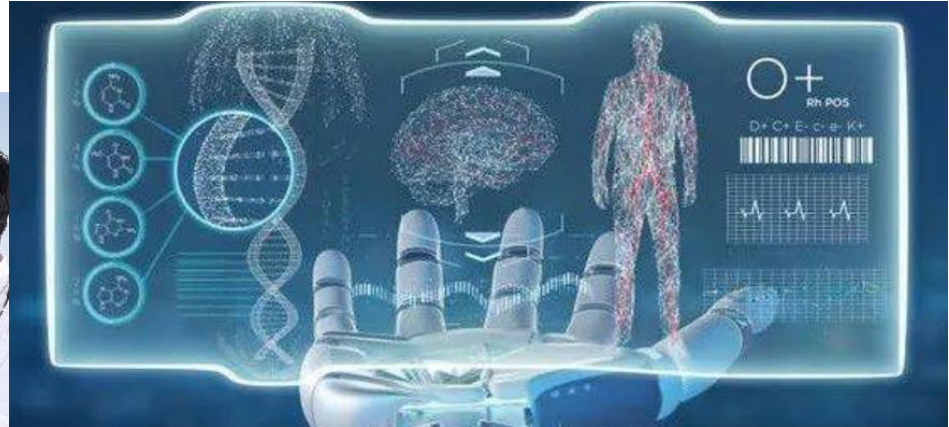
¹ CSIRO A&F, ² ANU, ³ Australian Plant Phenomics Facility, ⁴ CSIRO Data61



❖ AI in our life



❖ AI in our life



Example AI in our current life



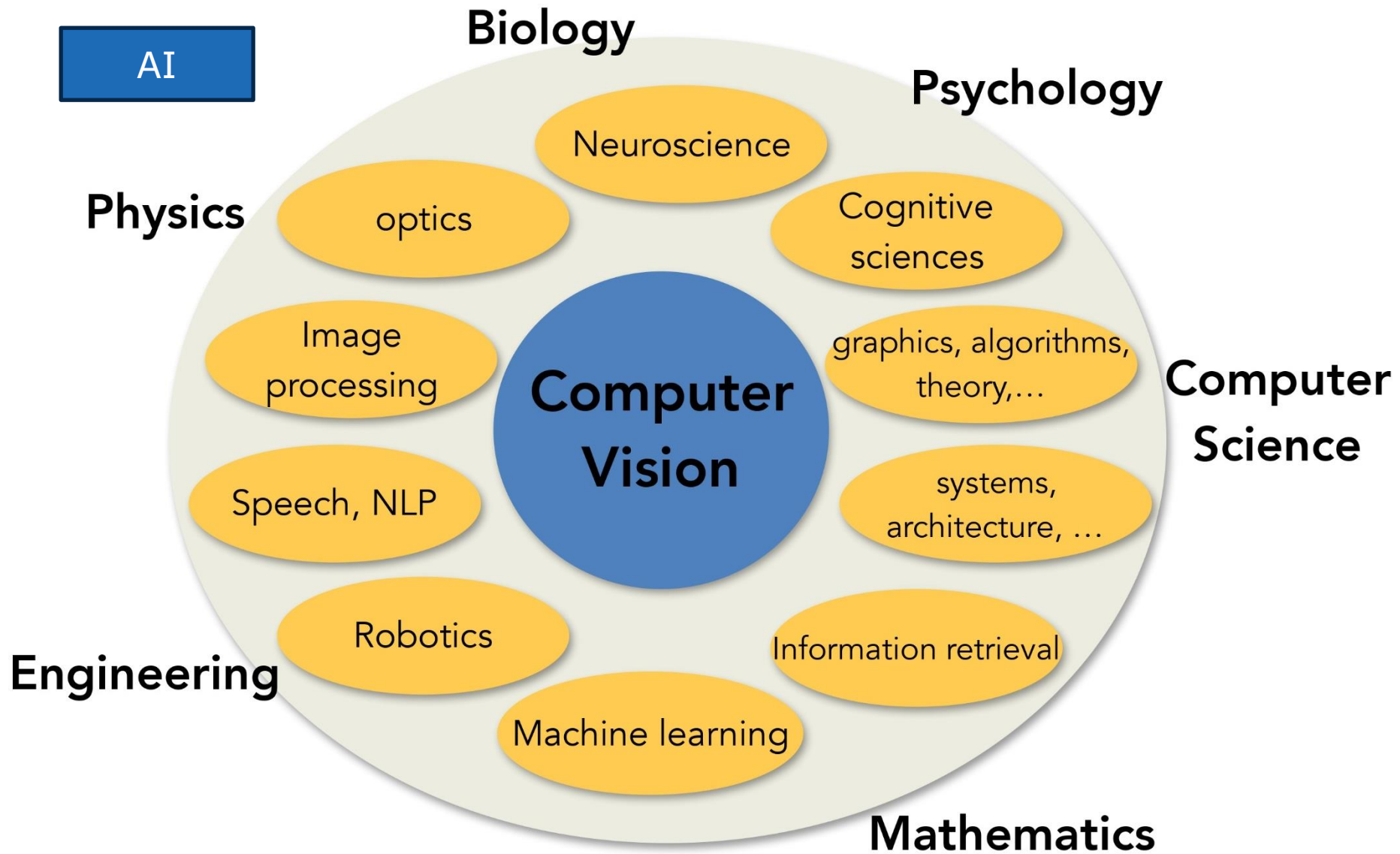
Example AI in our current life



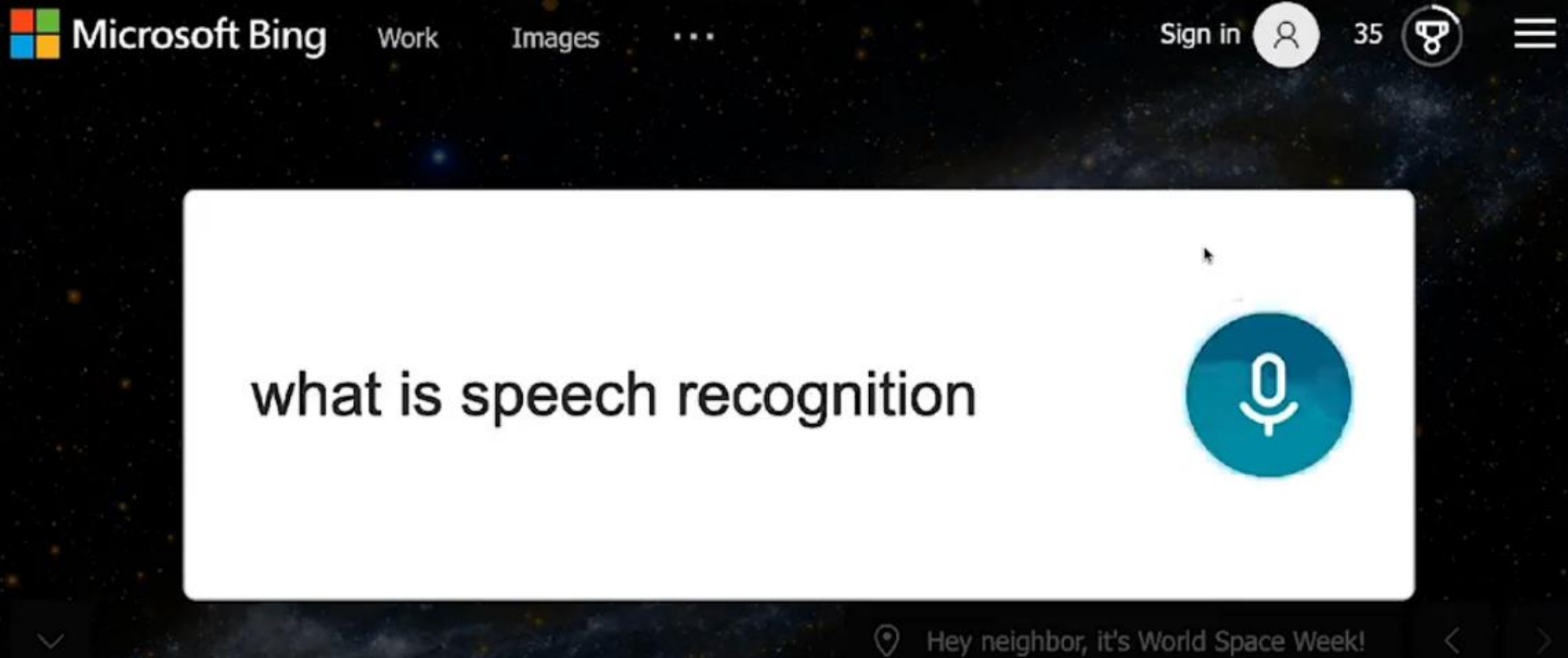
Example AI in our current life



AI



Intro

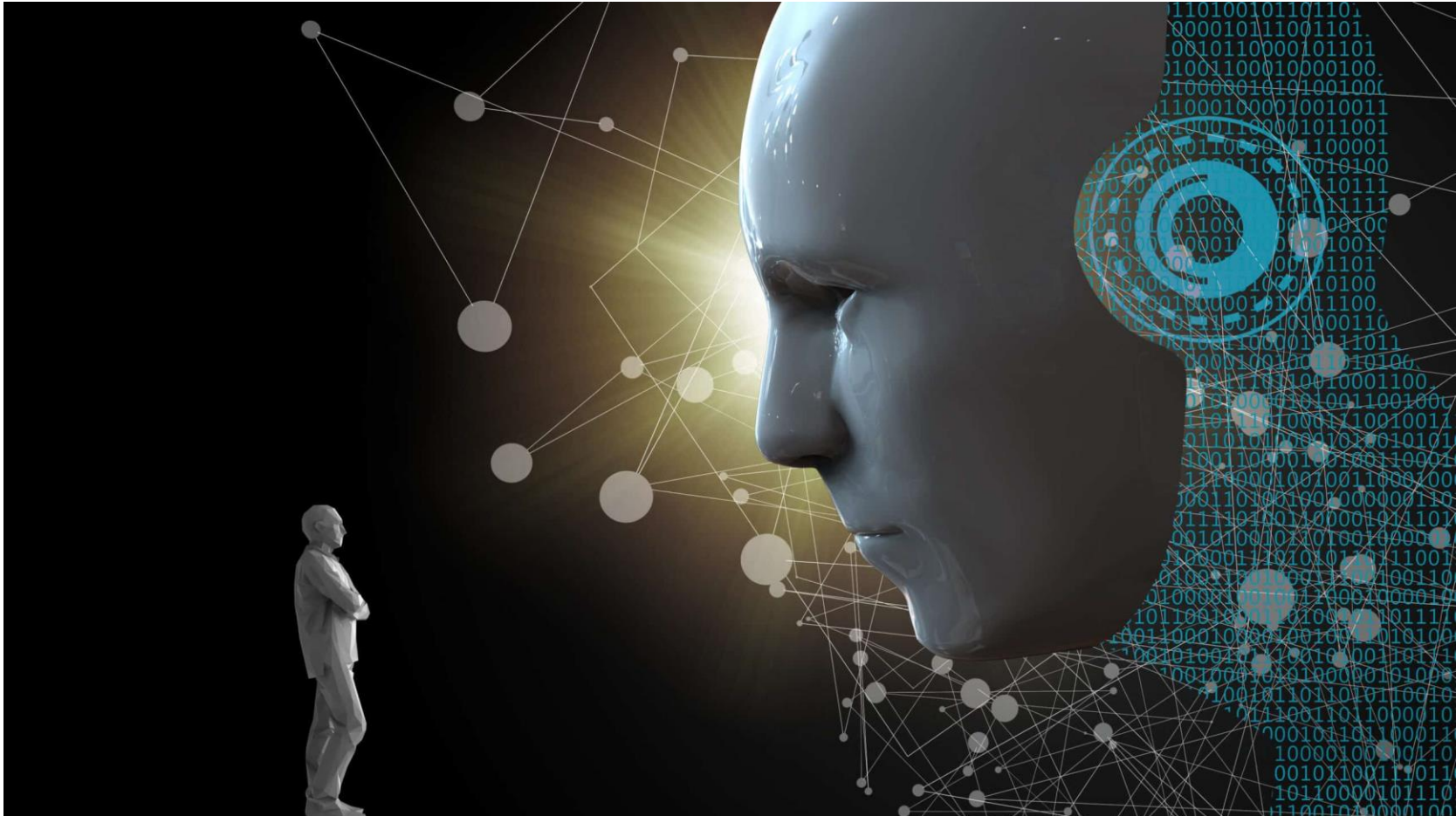


Intro



AI is always developing

Artificial Intelligence → **Intelligence**



Part I : Philosophy in AI

What is Intelligence, anyway?



R. J. Sternberg: *“Viewed narrowly, there seem to be almost as many definitions of intelligence as there were experts asked to define it.”*

It is useful to think of intelligence in terms of an open collection of attributes.

Characteristics of Intelligence(1)

❖ *Perception*



Manipulation, integration, and interpretation of data provided by sensors, including purposeful, goal-directed, active perception



■ *Action*

Coordination, control, and use of effectors to accomplish a variety of tasks, including exploration and manipulation of the environment, including design and construction of tools towards this end.

Characteristics of Intelligence(2)



- ***Reasoning***

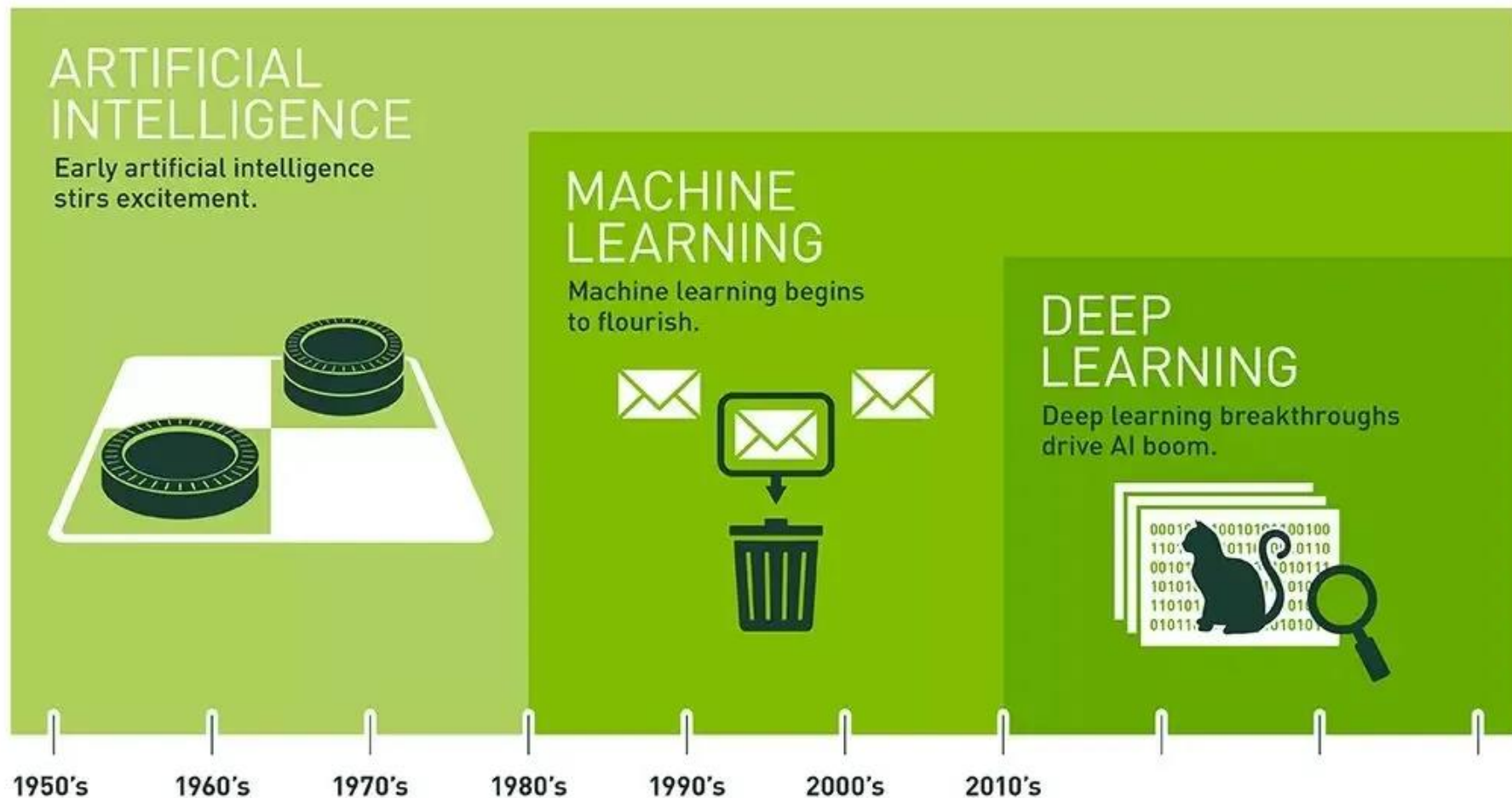
Deductive (logical) inference, inductive inference, analogical inference, hypothetical reasoning, ..., including reasoning in the face of uncertainty and incomplete information.

- ***Problem-solving***

Setting of goals (without explicit instructions from another entity), Formulation of plans, Evaluating and choosing among alternative plans, adapting plans in the face of unexpected changes

Part II: Ideas for AI

AI ML DL

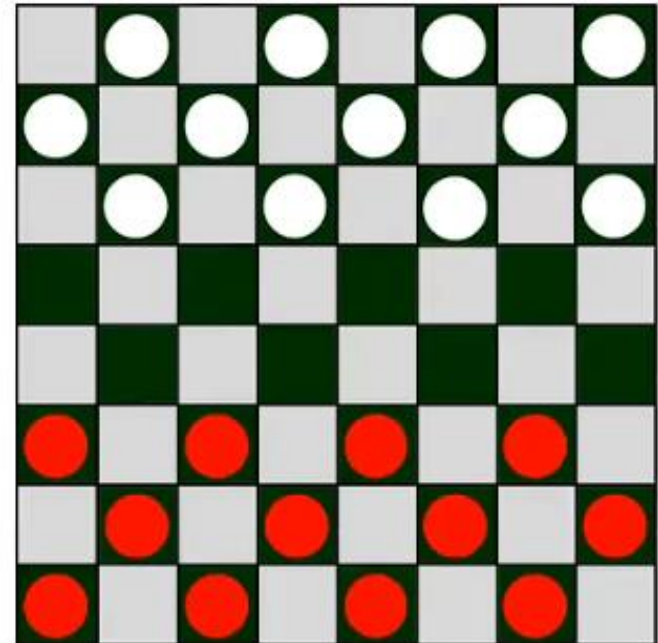


Since an early flush of optimism in the 1950s, smaller subsets of artificial intelligence – first machine learning, then deep learning, a subset of machine learning – have created ever larger disruptions.

Machine learning

“Field of study that gives computers the ability to learn without being explicitly programmed.”

Arthur Samuel (1959)



If the checkers program had been allowed to play only ten games (instead of tens of thousands) against itself, a much smaller number of games, how would this have affected its performance?

- ☐ Would have made it better
- ☐ Would have made it worse



Kinds of Learning

❖ **Supervised Learning**

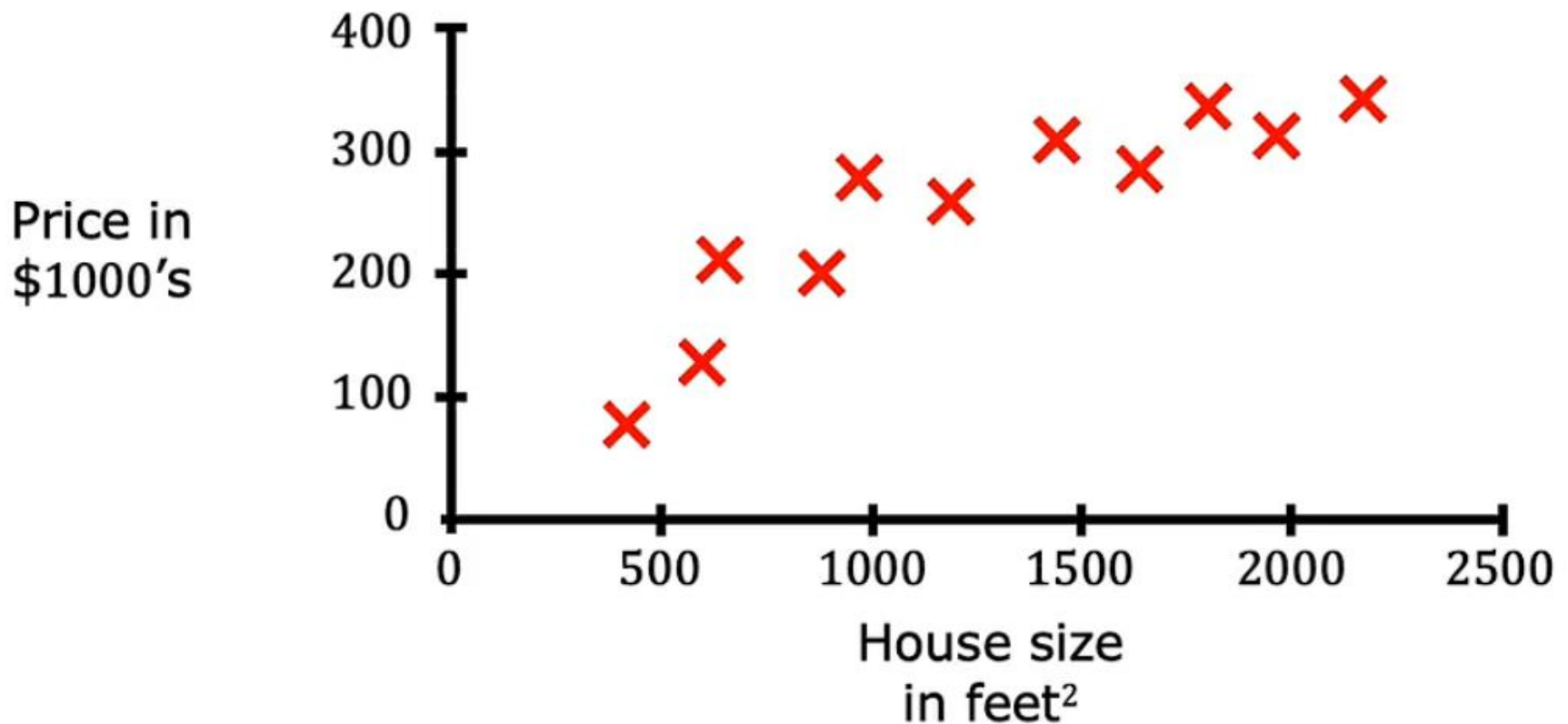
Given a set of example input/output pairs, find a rule that does a good job of predicting the output associated with a new input.

Training data, Validation data, Test data

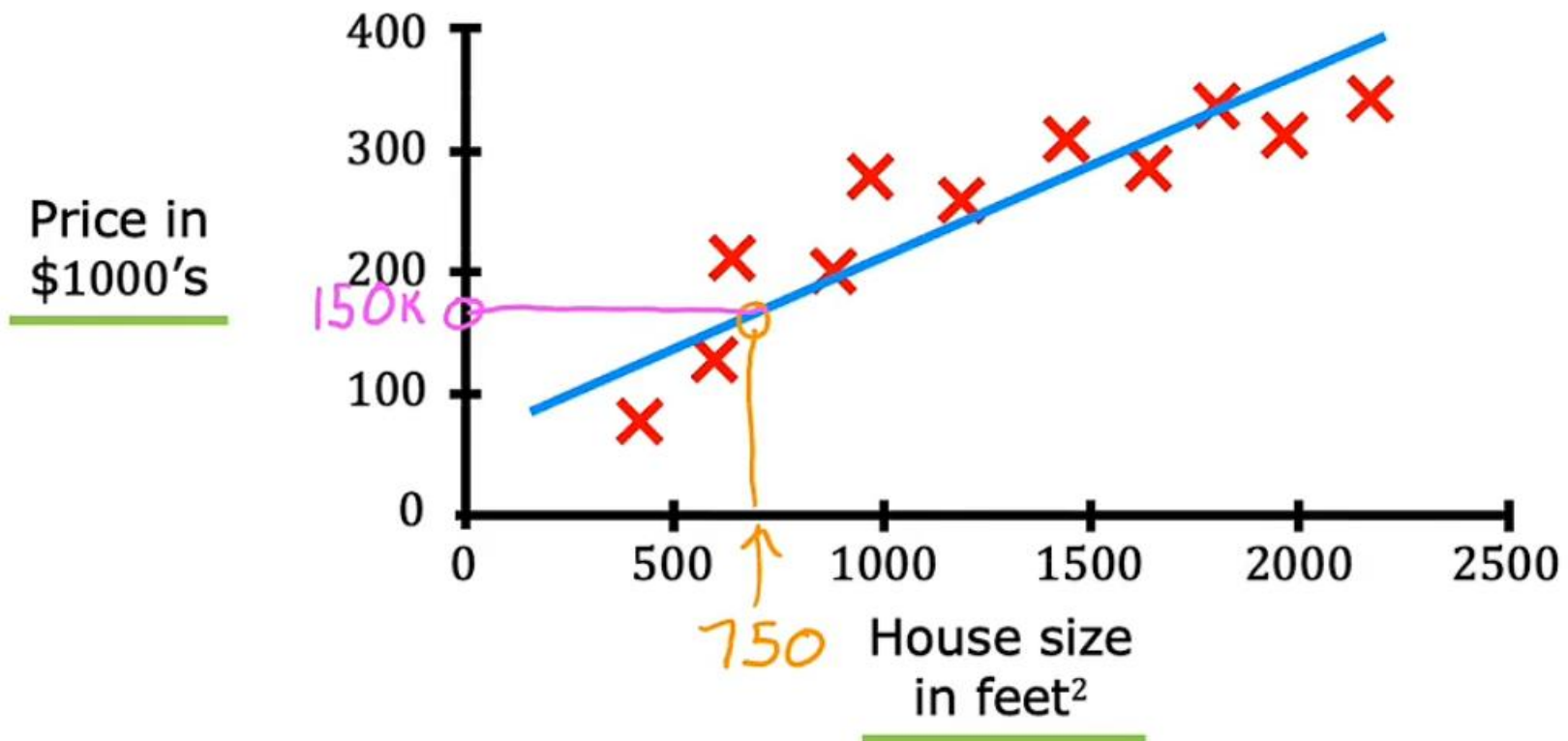
Kinds of Learning

Input (X)		Output (Y)	Application
email	→	spam? (0/1)	spam filtering
audio	→	text transcripts	speech recognition
English	→	Spanish	machine translation
ad, user info	→	click? (0/1)	online advertising
image, radar info	→	position of other cars	self-driving car

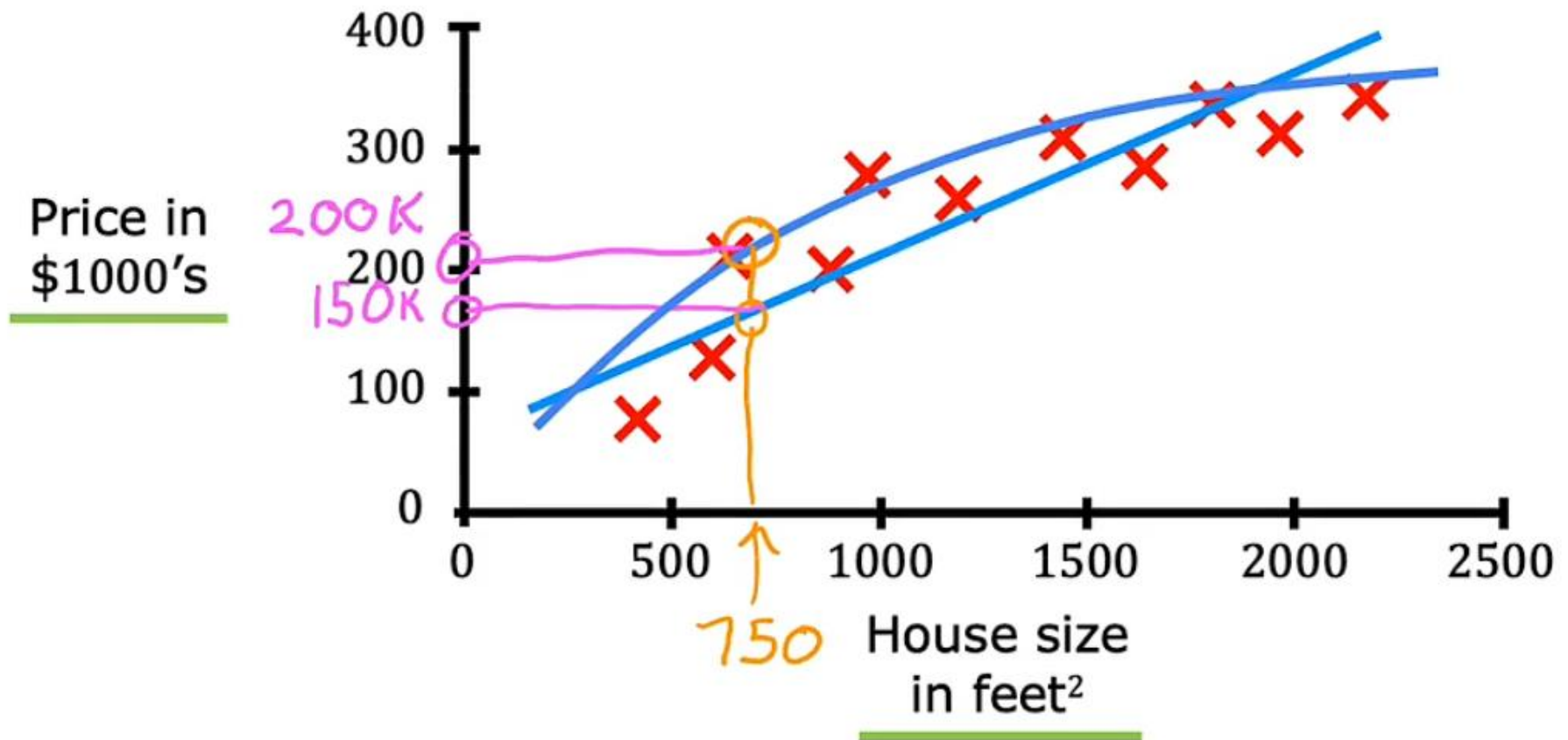
Regression: Housing price prediction



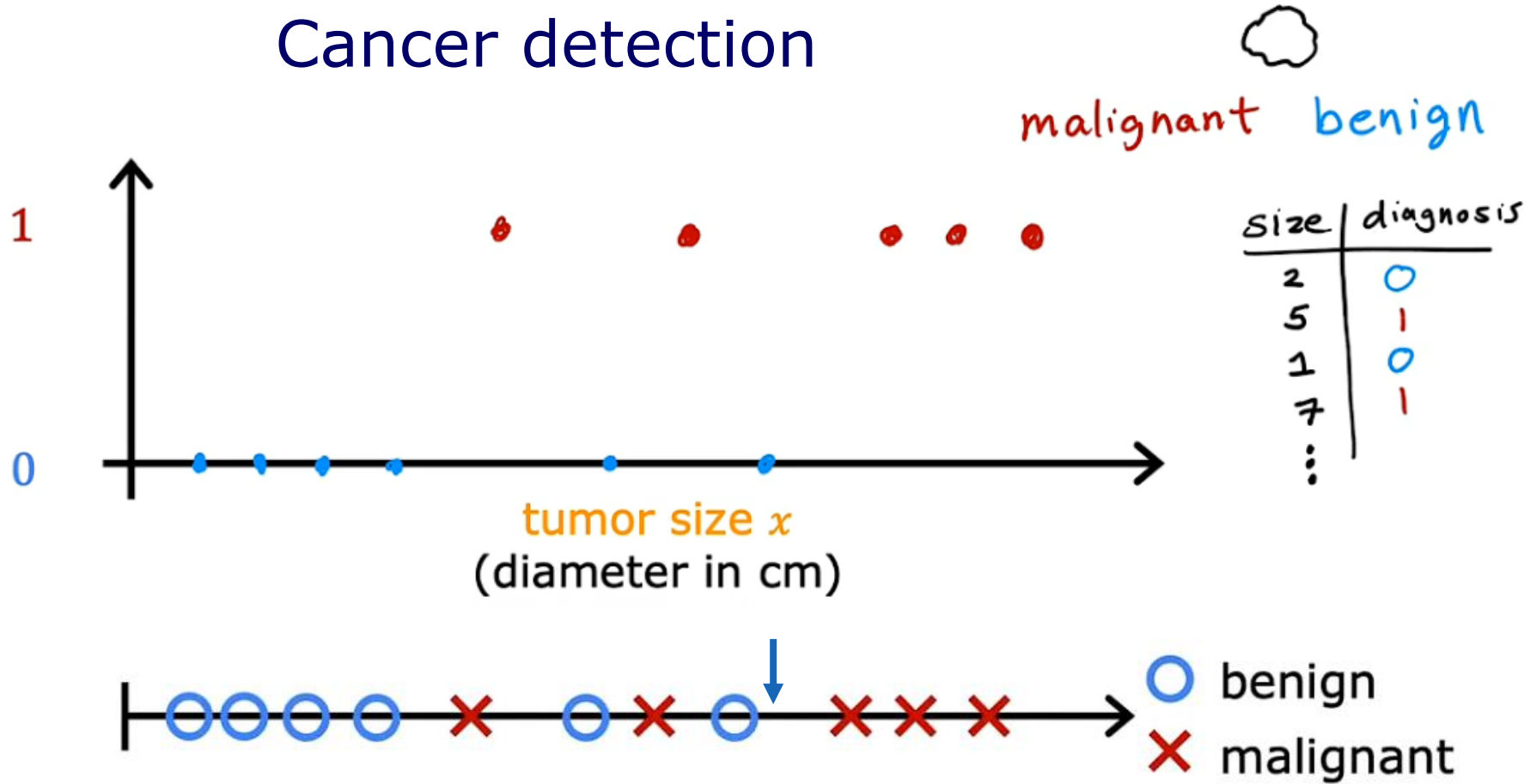
Regression: Housing price prediction



Regression: Housing price prediction



Cancer detection



- benign
- ✗ malignant *type 1*
- △ malignant type 2



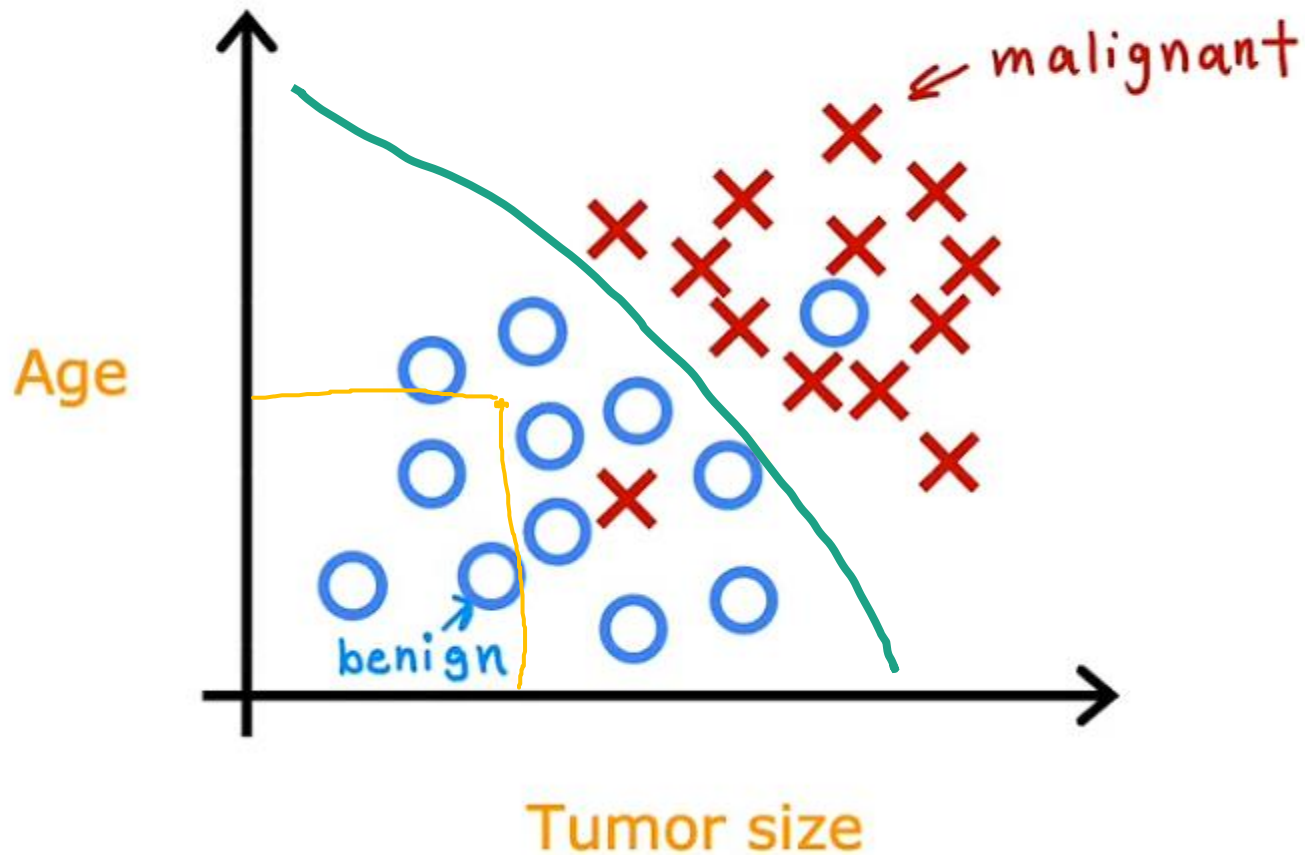
class category

Classification

predict **categories** *cat dog benign malignant 0, 1, 2*

small number of possible outputs

Two or more inputs



Supervised learning

Learns from being given “right answers”

Regression

Predict a number

infinitely many possible outputs

Classification

predict categories

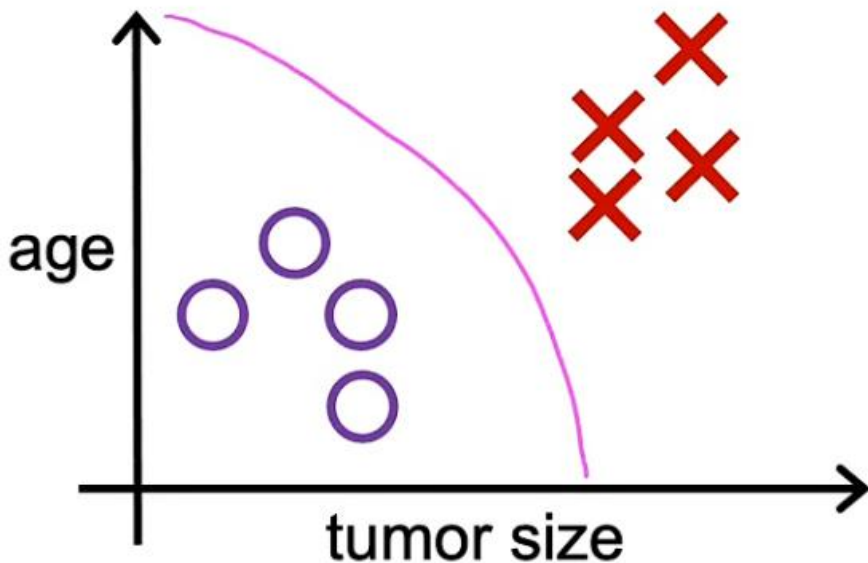
small number of possible outputs

Kinds of Learning

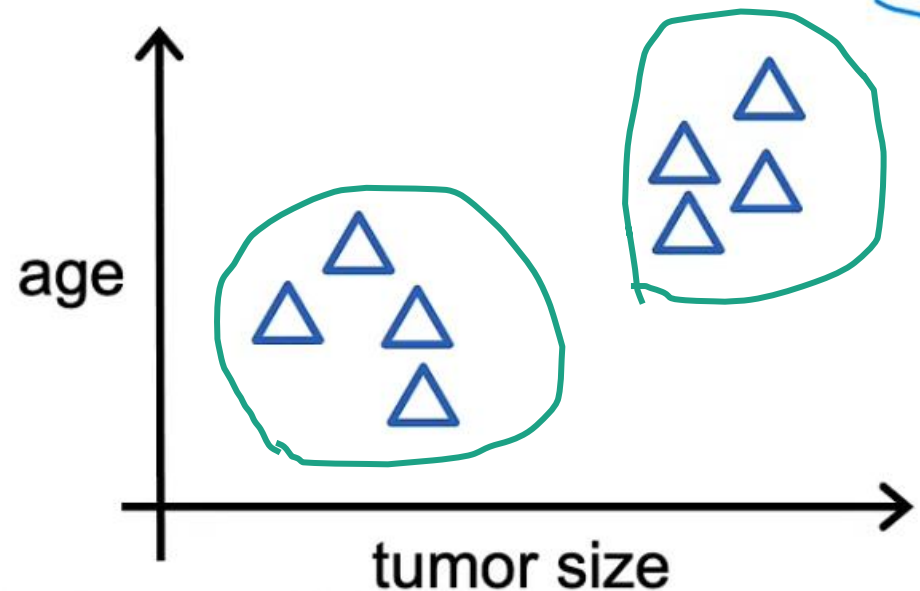
❖ **Unsupervised Learning (clustering)**

Given a set of examples, no labeling of them, group them into 'natural' clusters.

Supervised learning
Learn from data **labeled**
with the "right answers"



Unsupervised learning
Find something interesting
in **unlabeled** data.



Clustering: Google news

Giant panda gives birth to rare twin cubs at Japan's oldest zoo

USA TODAY · 6 hours ago



• Giant panda gives birth to twin cubs at Japan's oldest zoo

CBS News · 7 hours ago

• Giant panda gives birth to twin cubs at Tokyo's Ueno Zoo

WHBL News · 16 hours ago

• A Joyful Surprise at Japan's Oldest Zoo: The Birth of Twin Pandas

The New York Times · 1 hour ago

• Twin Panda Cubs Born at Tokyo's Ueno Zoo

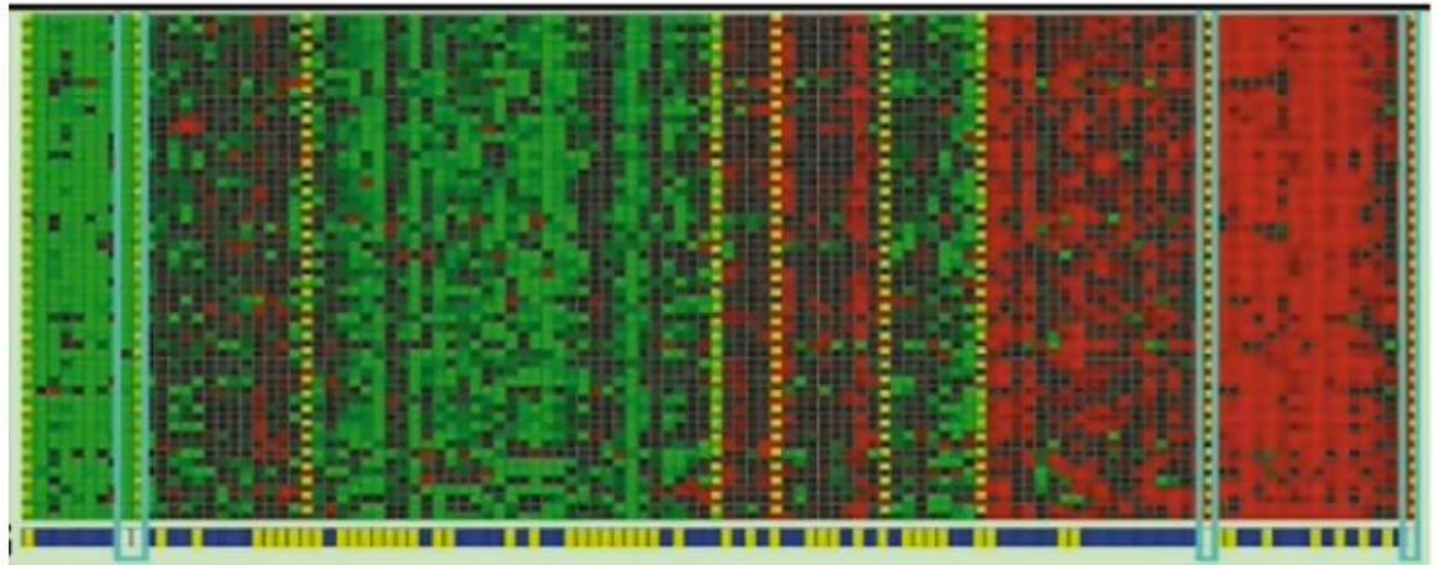
PEOPLE · 6 hours ago

 [View Full Coverage](#)



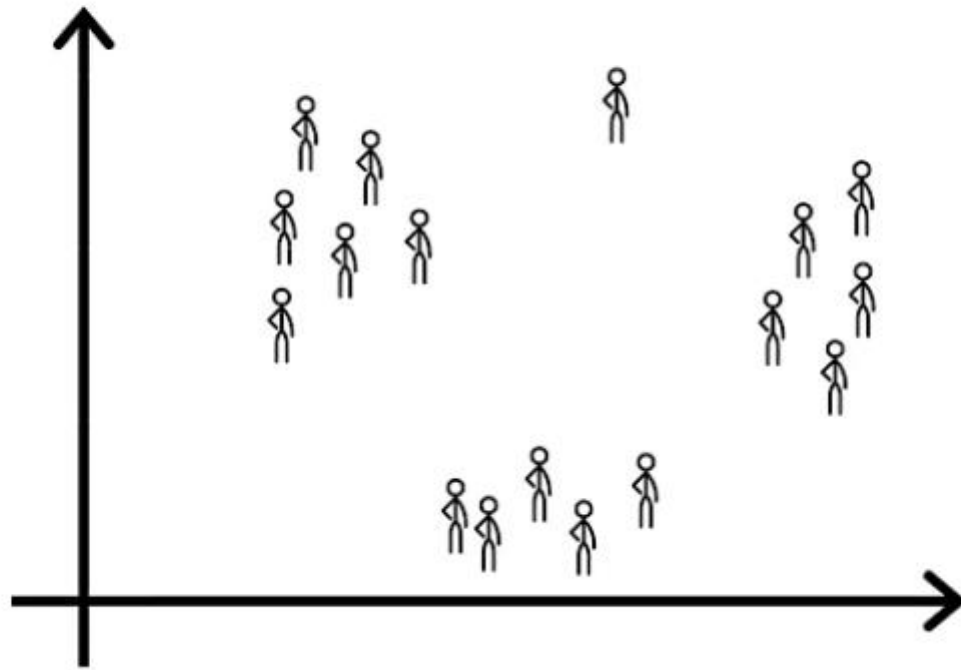
Clustering: DNA microarray

genes
(each row)

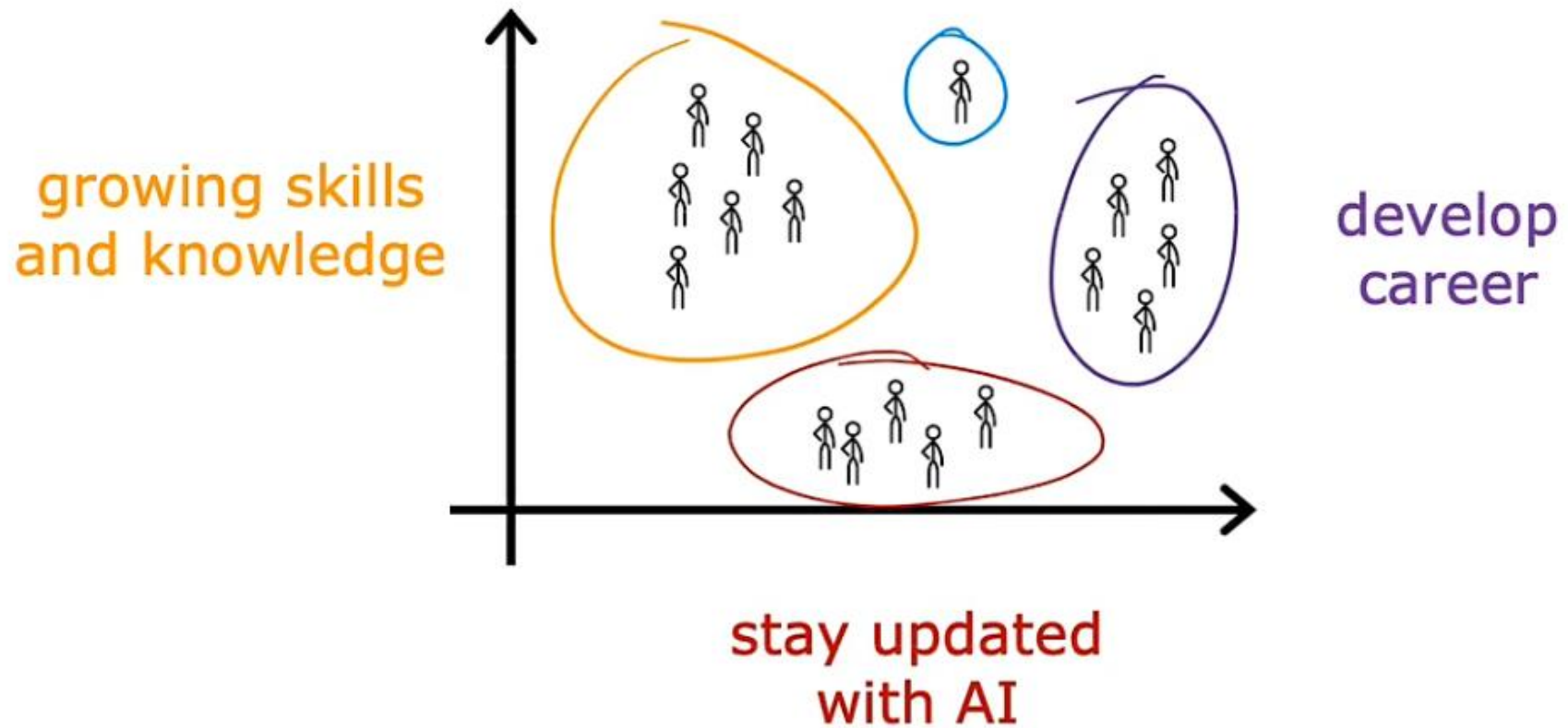


individuals
(each column)

Clustering: Grouping customers



Clustering: Grouping customers



Unsupervised learning

Data only comes with inputs x , but not output labels y .
Algorithm has to find **structure** in the data.

Clustering

Group similar data points together.

Dimensionality reduction

Compress data using fewer numbers.

Anomaly detection

Find unusual data points.

Question

Of the following examples, which would you address using an **unsupervised** learning algorithm?

- ☐ Given email labeled as spam/not spam, learn a spam filter.
- ☐ Given a set of news articles found on the web, group them into sets of articles about the same story.
- ☐ Given a database of customer data, automatically discover market segments and group customers into different market segments.
- ☐ Given a dataset of patients diagnosed as either having diabetes or not, learn to classify new patients as having diabetes or not

Kinds of Learning contd.

❖ **Semi-supervised Learning**

Combination of supervised and unsupervised learning.

Kinds of Learning contd.

❖ **Reinforcement Learning**

An agent interacting with the world makes observation, takes actions, and is rewarded or punished; it should learn to choose actions in such a way as to obtain a lot of reward.

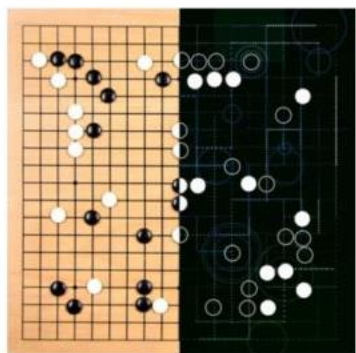
Kinds of Learning contd.

Our reinforcement learning agent is Mario, who will learn to play on its own



AlphaGo - Ground-breaking

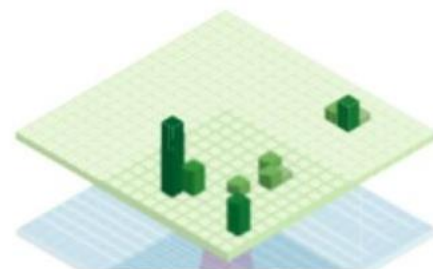




棋盘信息



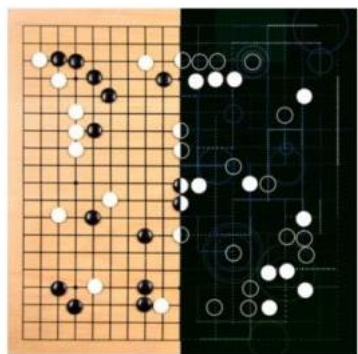
神经网络



落子选择

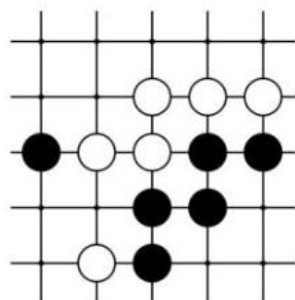
模块：

- 策略网络 (Policy Network) , 给定当前局面, 预测并采样下一步的走棋;
- 价值网络 (Value Network) , 给定当前局面, 估计是白胜概率大还是黑胜概率大;
- 蒙特卡洛树搜索 (Monte Carlo Tree Search) , 阅读未来的棋子移动的序列。



棋盘信息

落子颜色Stone colour



```
[0 0 0 0 0]
[0 0 1 1 1]
[0 1 1 0 0]
[0 0 0 0 0]
[0 1 0 0 0]
```

自己棋子
(这里是白棋)

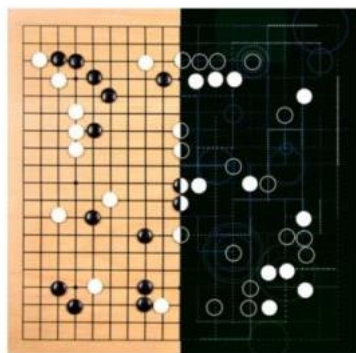
```
[0 0 0 0 0]
[0 0 0 0 0]
[1 0 0 1 1]
[0 0 1 1 0]
[0 0 1 0 0]
```

对方棋子
(这里是黑棋)

```
[1 1 1 1 1]
[1 1 0 0 0]
[0 0 0 0 0]
[1 1 0 0 1]
[1 0 0 1 1]
```

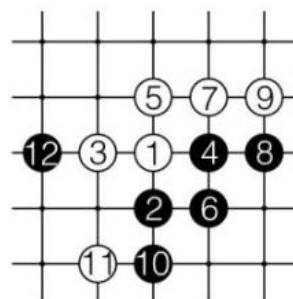
空白位置

轮次Turn since



棋盘信息

One Hot encoding 二值化



[0 0 0 0 0]
[0 0 0 0 0]
[1 0 0 0 0]
[0 0 0 0 0]
[0 0 0 0 0]

P1

[0 0 0 0 0]
[0 0 0 0 0]
[0 0 0 0 1]
[0 0 0 0 0]
[0 0 0 0 0]

P5

[0 0 0 0 0]
[0 0 0 0 0]
[0 0 0 0 0]
[0 0 0 0 0]
[0 1 0 0 0]

P2

[0 0 0 0 0]
[0 0 0 1 0]
[0 0 0 0 0]
[0 0 0 0 0]
[0 0 0 0 0]

P6

[0 0 0 0 0]
[0 0 0 0 0]
[0 0 0 0 0]
[0 0 0 0 0]
[0 0 1 0 0]

P3

[0 0 0 0 0]
[0 0 0 0 0]
[0 0 0 0 0]
[0 0 0 1 0]
[0 0 0 0 0]

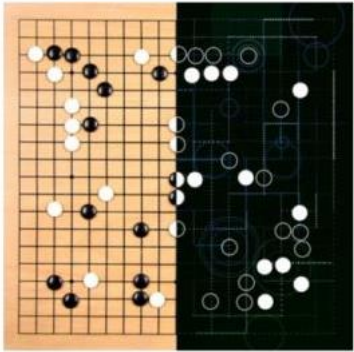
P7

[0 0 0 0 0]
[0 0 0 0 1]
[0 0 0 0 0]
[0 0 0 0 0]
[0 0 0 0 0]

P4

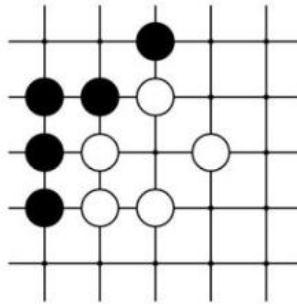
[0 0 0 0 0]
[0 0 1 0 0]
[0 1 1 1 0]
[0 0 1 0 0]
[0 0 0 0 0]

P8



棋盘信息

合法性Sensibleness



```
[1 1 0 1 1]
[0 0 0 1 1]
[0 0 0 0 1]
[0 0 0 1 1]
[1 1 1 1 1]
```

Learning issues

- ❖ **Overfitting (generalization ability)** : can the machine well-trained on observed data behave well on other data either?
- ❖ **Bias**: which hypotheses are preferred?
- ❖ **Robustness**: how does the training data influence the learning result? Data Scale, Change, Noise, and Imbalance
- ❖ **Transparency**: can we understand what and how has been learnt?
- ❖ **Computation Complexity**: what is the efficiency of the learning algorithms? Time, Memory, Scalability, convergency

After class reading

Connectionism

❖ A brief history

M-P neuron (McCulloch & Pitts) → Perceptron (Rosenblatt) → Hopfield Model, B-P Learning Method (Rumelhart & McClelland) → Deep Learning (Geoffrey Hinton)

❖ Applications

Recognition, Vision, Business, Medical,

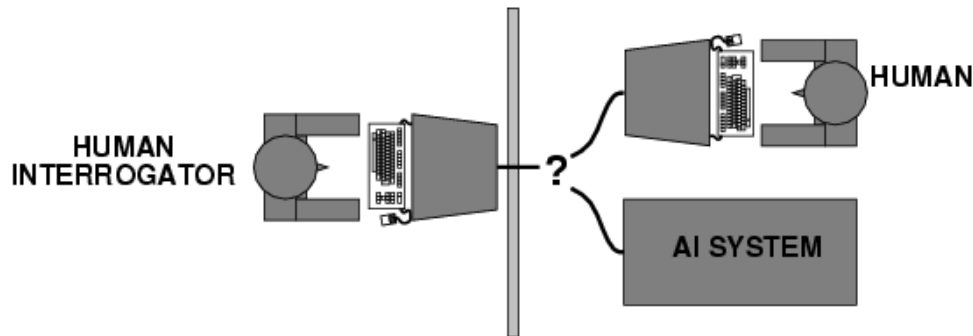
❖ Core Issues

- *Topology*
- *Learning Methods*

Turing test

❖ When does a system behave intelligently?

- A. M. Turing (1950) Computing Machinery and Intelligence. Mind 49: 433-460.
- *Operational test of intelligence*: imitation game



- Requires the collaboration of major components of AI: knowledge, reasoning, language understanding, learning, ...

Chinese Room Argument

A man is in a room with a book of rules. Chinese sentences are passed under the door to him. The man looks up in his book of rules how to process the sentences. Eventually the rules tell him to copy some Chinese characters onto paper and pass the resulting Chinese sentences as a reply to the message he has received. The dialog continues. To follow these rules the man need not understand Chinese.

Searle, John. R. (1980) Minds, brains, and programs. *Behavioral and Brain Sciences* 3 (3): 417-457



❖ **Therefore, Searle says:**
- the idea of a non-biological machine being intelligent is incoherent

End of Lecture1

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